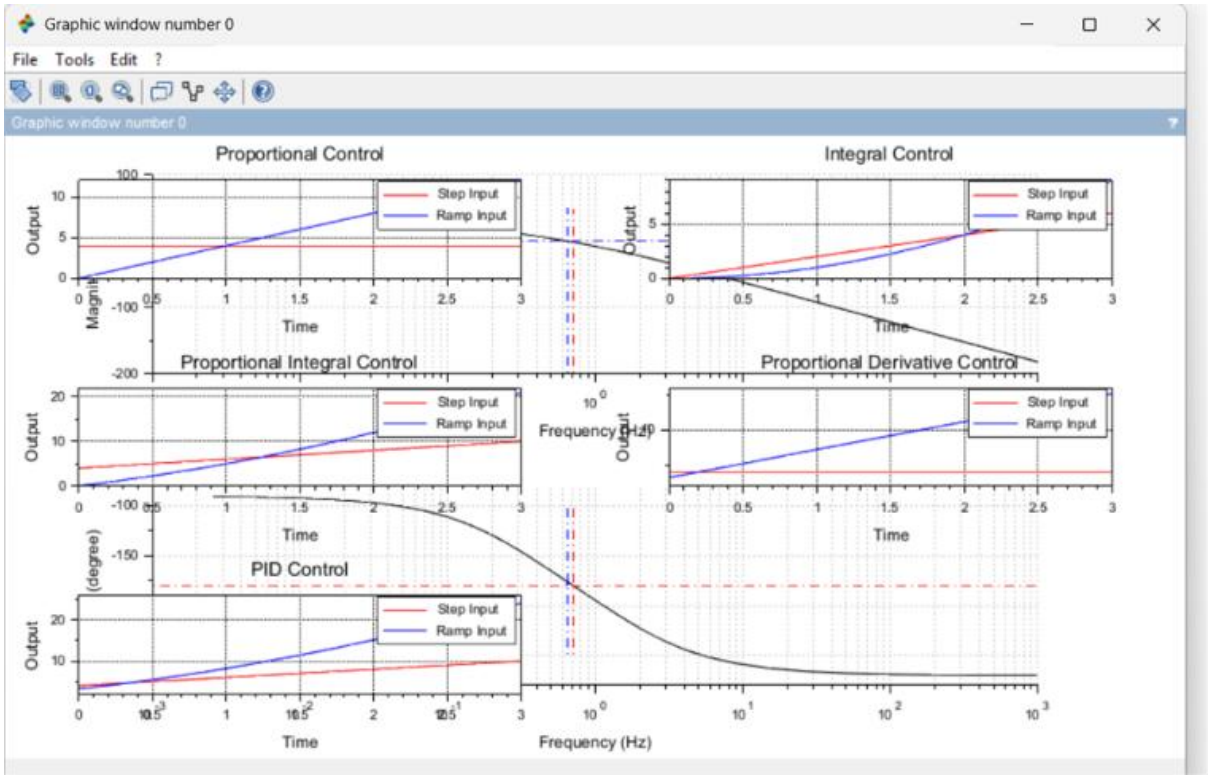
PROGRAM CODE :

```
clc;
clear;
xdel(winsid());
Kp=4;//proportional gain
Ki1=2;//integral gain
Td=0.8;//differential time
Ti=2;//integral time
Ki2=Kp/Ti;
s=%s;
Gi=syslin('c',Ki1/s);
t=0:0.05:3;
ramp=t;
subplot(321);
p1=Kp*ones(1,length(t));
p2=Kp*t;
plot2d(t,p1,style=2);
plot2d(t,p2,style=3);
xtitle('Proportional control','t','y');
legend('step input','ramp input');
xgrid(color('gray'));
subplot(322);

i1=csim("step",t,Gi);
i2=csim(ramp,t,Gi);
plot2d(t,i1,style=2);
plot2d(t,i2,style=3);
xtitle('Integral control','t','y');
```

```
xgrid(color('gray'));
i1=i1*Ki2/Ki1;//change of gain
i2=i2*Ki2/Ki1;
subplot(323);
plot2d(t,p1+i1,style=2);
plot2d(t,p2+i2,style=3);
xtitle('Proportional integral control','t','y');
xgrid(color('gray'));
subplot(324);
pd1=p1;
pd2=p2+Kp*Td*ones(1,length(t));//derivative term
plot2d(t,pd1,style=2);
plot2d(t,pd2,style=3);
xtitle('Proportional plus derivative control','t','y');
xgrid(color('gray'));
subplot(325);
plot2d(t,pd1+i1,style=2);
plot2d(t,pd2+i2,style=3);
xtitle('P.I.D. control','t','y');
xgrid(color('gray'));
```

OUTPUT :


Graphic window number 0
File Tools Edit ?
Graphic window number 0
Proportional Control
Integral Control
Proportional Integral Control
Proportional Derivative Control
PID Control
Step Input
Ramp Input
Output
Time
Magnitude (degree)
Frequency (Hz)